HCC3 EXAM 4

PART 1 • LABOR & BIRTH • FETAL MONITORING • COMPLICATIONS • OBSTETRIC EMERGENCIES

Complete, colorful nursing study guide organized in the exact order of your blueprint. Includes detailed explanations, nursing priorities, medication precautions, emergency actions, and 12 practice questions with rationales.

1. LABOR AND BIRTH PROCESS

The Five P's — factors affecting labor

P	Meaning and clinical importance
Passenger	Fetus and placenta: fetal size, lie, presentation, position, attitude, and station determine how the fetus travels through the pelvis.
Passageway	Maternal bony pelvis, cervix, vagina, and pelvic floor. Shape and dimensions affect engagement and descent.
Powers	Primary: involuntary uterine contractions cause effacement/dilation. Secondary: maternal pushing assists fetal expulsion.
Position	Side-lying improves perfusion; upright positions promote descent; hands-and-knees may relieve back labor; supine positioning may cause aortocaval compression.
Psyche	Fear, stress, fatigue, trauma history, expectations, and support influence coping and possibly contraction effectiveness.

Passenger: terminology to know

- Lie: longitudinal (parallel), transverse (perpendicular), oblique (angled). Persistent transverse lie generally prevents vaginal birth.
- Presentation: cephalic/head-first (usually favorable); breech/buttocks or feet (higher cord-prolapse risk); shoulder (often transverse lie).
- Attitude: flexion (chin tucked, smaller presenting head diameter) versus extension (larger diameter).
- Position: LOA and ROA are generally favorable; occiput posterior (OP) may cause back labor and prolonged labor; occiput transverse may need rotation.
- Station: presenting part relative to ischial spines: negative above (−1 to −5), 0 at spines, positive below (+1 to +5). Engagement occurs when biparietal diameter enters pelvic inlet.

Passageway and powers

- Pelvis: gynecoid rounded/favorable; android heart-shaped; anthropoid oval anteroposteriorly; platypelloid flattened. These are descriptive types, not definitive delivery predictions.
- Dilation: cervix opens 0–10 cm. Effacement: thins 0–100%. Example 5 cm/80%/−1 = 5 cm open, 80% thin, presenting part above spines.

- Contraction frequency: beginning of one to beginning of next; duration: beginning to end; intensity: strength; resting tone: relaxation between contractions.
- Uteroplacental blood flow temporarily falls during contractions; relaxation between contractions is important for fetal oxygenation.

Signs preceding labor and onset

- Lightening: fetal descent into pelvis, easier breathing but increased pelvic pressure.
- Braxton Hicks: irregular contractions without progressive cervical change.
- Bloody show/mucus plug: blood-tinged mucus or mucus loss with cervical change.
- Nesting, increased pelvic pressure, cervical softening and ripening may occur before labor.
- True labor: progressively stronger, more regular contractions with cervical dilation/effacement; usually persist despite rest. False labor: irregular, may diminish with rest/hydration, no progressive cervical change.
- Labor initiation involves prostaglandins, increased oxytocin responsiveness, functional progesterone withdrawal, estrogen effects, and fetal/placental signaling.

EXAM PRIORITY: Progressive cervical change is the most definitive distinction between true and false labor.

Four stages of labor

Stage	Clinical changes	Priority nursing care
1: onset → 10 cm	Latent: 0 to about 6 cm, gradual dilation; active: about 6–10 cm, stronger contractions. Transition is traditional term near 8–10 cm; nausea, shaking, rectal pressure, irritability may occur.	Monitor maternal vitals, FHR, contractions, pain, progress; encourage voiding, rest, hydration as permitted, mobility and coping. Assess urge to push before encouraging pushing.
2: 10 cm → infant	Fetal descent, pushing, crowning and birth.	FHR surveillance, effective pushing, positioning, rest, delivery setup, neonatal resuscitation readiness; recognize dystocia.
3: infant → placenta	Placenta separates: blood gush, cord lengthening, firm globular rising uterus.	Monitor bleeding and tone; prophylactic oxytocin as ordered; observe placental completeness. Never forcibly pull cord.
4: early recovery	Generally first 1–2 postpartum hours.	Fundus firm/midline, lochia, vitals, bladder, perineum, pain, anesthesia recovery, bonding/feeding.

Seven cardinal movements of labor

- 1. Engagement: biparietal diameter passes pelvic inlet.
- 2. Descent: presenting part moves downward; occurs throughout labor.
- 3. Flexion: chin toward chest reduces presenting head diameter.
- 4. Internal rotation: head aligns with pelvic outlet.
- 5. Extension: head extends beneath pubic symphysis and is born.
- 6. Restitution/external rotation: head realigns with shoulders, shoulders rotate.
- 7. Expulsion: shoulders and body deliver.

Fetal adaptations

- Contractions briefly reduce placental blood flow; fetal heart rate and circulation adapt to oxygen supply. Moderate variability generally reassures about current fetal status.
- Lung fluid clearance begins before and during labor; catecholamine surge supports newborn respiratory transition.
- Head molding is temporary skull-bone overlap. Caput succedaneum is scalp edema crossing sutures; cephalohematoma is subperiosteal blood collection that does not cross sutures.
- Persistent oxygen deficit may cause acidemia and abnormal FHR: recurrent late decelerations, persistent bradycardia, absent variability with recurrent decelerations, or sinusoidal pattern.

Maternal adaptations

- Cardiovascular: increased cardiac output during contractions and transient BP rise.
- Respiratory: higher oxygen demand; anxiety/pain may cause hyperventilation.
- GI: nausea/vomiting, slower gastric emptying. Urinary: bladder distention can obstruct descent; epidural may cause retention.
- Musculoskeletal: fatigue, back/pelvic pressure. Psychological: excitement, fear, increased focus, exhaustion.
- Report persistent hypotension, fever, hypoxemia, excessive bleeding, or altered mental status.

2. MAXIMIZING COMFORT

Visceral pain (mostly stage 1)	Somatic pain (mostly stage 2)
Uterine contractions, cervical dilation and ischemia; dull, diffuse lower abdominal/back cramps; pathways T10–L1.	Stretching of vagina, perineum and pelvic floor; sharper, localized perineal/rectal pain; primarily S2–S4 via pudendal nerve.

Pain perception, expression, and influencing factors

- Pain is subjective: patients may cry, remain silent, vocalize, move, withdraw, or request medication. Never judge pain severity by expression alone.
- Influences: fear/anxiety, previous birth, trauma, culture, fatigue, fetal OP position, labor length/intensity, support, expectations, and sense of control.
- Nursing care: ask about pain and desired relief, explain options, support informed consent and continuous companionship.

Nonpharmacological measures

- Breathing patterns, relaxation, guided imagery, music, mindfulness, continuous labor support.
- Massage, effleurage, sacral counterpressure (especially OP/back labor), warm/cold packs.
- Frequent position changes, upright movement, birthing ball, pelvic rocking, hands-and-knees, hydrotherapy, TENS when available.

Sedatives, systemic analgesics, and anesthesia

Category	Examples / purpose	Nursing cautions
Sedatives	Selected sedating drugs reduce anxiety/drowsiness; not contraction analgesia or labor augmentation.	Maternal sedation, hypotension, respiratory depression; neonatal sedation near birth. Monitor RR, alertness, BP, fetus as appropriate.
Systemic opioids	Fentanyl, butorphanol, nalbuphine; sometimes morphine. Reduce pain but cross placenta.	Monitor maternal RR, SpO2, sedation, BP, FHR and timing of delivery. Risk neonatal respiratory depression near birth. Naloxone is not a substitute for newborn ventilation.
Epidural	Local anesthetic ± opioid in epidural space; sustained analgesia.	Check baseline FHR/BP, IV access, consent; then frequent BP/FHR, block, respirations, bladder, motor function. Hypotension can impair placental perfusion.
Spinal / combined	Rapid spinal block; combined spinal-epidural offers rapid and sustained relief.	Watch hypotension, block height, respiratory compromise.
Pudendal/local	Pudendal blocks lower vagina/perineum; local infiltration for repair.	Does not relieve uterine contraction pain.
General anesthesia	Unconsciousness for selected urgent operations.	Airway/aspiration risks, maternal and neonatal effects.

EXAM PRIORITY: After epidural-related hypotension: reposition laterally, assess FHR and BP, give ordered fluids/vasopressor (phenylephrine or ephedrine), notify anesthesia. High spinal signs (breathing difficulty, rising block, severe hypotension) require immediate emergency response.

Medications used to augment labor

Oxytocin stimulates uterine contractions after spontaneous labor has begun; amniotomy may also augment labor. Sedatives, opioids and nerve blocks do NOT augment labor. Full oxytocin details appear in Section 12.

3. FETAL ASSESSMENT DURING LABOR

Purpose, physiological basis, and monitoring techniques

- FHR monitoring assesses fetal response to contractions and identifies patterns suggesting possible inadequate oxygenation; it cannot directly diagnose fetal acidemia.
- Baseline FHR normal 110–160 bpm (10-minute average excluding accelerations/decelerations and marked variability). Tachycardia >160 for ≥10 minutes; bradycardia <110 for ≥10 minutes.
- Tachycardia causes: fever, infection, medications, hypoxemia. Bradycardia causes: hypotension, cord compression, hypoxemia, other emergencies.
- Intermittent auscultation: Doppler/fetoscope, often appropriate for low-risk labor.
- External EFM: ultrasound transducer for FHR, TOCO for contraction timing/duration; external TOCO does NOT reliably measure intensity or resting tone.
- Internal fetal scalp electrode: accurate FHR with ruptured membranes and appropriate cervical access; avoid with certain infections/conditions.

- IUPC: measures contraction intensity/resting tone after rupture; MVUs quantify uterine activity. Around 200 MVUs/10 min often considered adequate in arrest evaluation.

Variability and accelerations

Pattern	Definition / significance
Absent variability	Undetectable amplitude; concerning especially with recurrent decelerations.
Minimal variability	Detectable amplitude ≤5 bpm; possible fetal sleep, drugs, prematurity, hypoxemia.
Moderate variability	6–25 bpm; reassuring.
Marked variability	>25 bpm; assess clinical context.
Acceleration ≥32 weeks	Rise ≥15 bpm for ≥15 seconds, <2 minutes.
Acceleration <32 weeks	Rise ≥10 bpm for ≥10 seconds.

Decelerations and nursing interventions

Pattern	Cause and tracing	Response
Early	Head compression; gradual mirror of contraction, nadir near contraction peak; generally benign.	Continue routine assessment.
Late	Uteroplacental insufficiency; gradual decrease whose nadir follows contraction peak; possible hypotension, excessive contractions, placental dysfunction.	Reposition laterally; stop/reduce oxytocin; assess BP and oxygenation; fluids if indicated; correct hypotension; notify provider; expedite delivery if unresolved.
Variable	Cord compression; abrupt fall ≥15 bpm lasting ≥15 sec and <2 min; timing varies.	Reposition, assess for cord prolapse when appropriate, reduce oxytocin if tachysystole, notify provider for recurrent patterns; ordered amnioinfusion may help.
Prolonged	Decrease ≥15 bpm for 2 to <10 min; ≥10 min becomes baseline change.	Immediate assessment/resuscitative measures; evaluate cord, abruption, hypotension, rupture and tachysystole; escalate urgently.

Three-tier fetal heart rate categories

Category	Interpretation and action
I	Baseline 110–160, moderate variability, no late or variable decelerations; early decelerations/accelerations may be present or absent. Routine care.
II	All indeterminate patterns not meeting I or III. Evaluate and correct reversible causes; reassess.
III	Absent variability WITH recurrent late or variable decelerations or bradycardia; OR sinusoidal pattern. Immediate evaluation/correction; expedite birth if unresolved.

- VEAL CHOP: Variable–Cord, Early–Head, Accelerations–Okay/reassuring, Late–Placental insufficiency.

- Tachysystole: >5 contractions in 10 minutes averaged over 30 minutes. Stop/reduce oxytocin, reposition, assess FHR/BP, notify provider, give ordered tocolytic such as terbutaline if indicated.
- Do not routinely give supplemental oxygen for FHR abnormalities when maternal oxygen saturation is normal; give oxygen for maternal hypoxemia or other indication.

EXAM PRIORITY: Oxytocin + tachysystole + recurrent late decelerations = STOP oxytocin and initiate corrective interventions.

4. NURSING CARE OF THE FAMILY DURING LABOR AND BIRTH

Admission assessment and diagnostics

- Assess gestational age, EDD, G/P, prenatal complications, allergies, medications, contractions, fetal movement, membrane status, bleeding, vital signs, FHR, GBS, support preferences.
- Prenatal labs: ABO/Rh, antibody screen, CBC (anemia/platelets), GBS, HIV, hepatitis B, syphilis, urine testing, glucose screening.
- Vaginal exam: dilation, effacement, station, presentation/position; avoid unnecessary exams after ROM.
- Leopold maneuvers: four abdominal palpations for fetal lie, presentation, position and engagement. Ultrasound confirms presentation/placenta/other findings.
- ROM testing: pooling, nitrazine, ferning, or amniotic-fluid protein tests; nitrazine false positives possible with blood/semen/alkaline contaminants.

Care by stage, supportive care, placenta

- Stage 1: maternal/fetal assessments, coping, comfort, fluids as allowed, bladder emptying, positioning, labor progress.
- Stage 2: pushing guidance, fetal surveillance, birth preparation, safety and neonatal readiness.
- Stage 3: observe placental separation, blood loss, uterine tone, give ordered prophylactic oxytocin, assess completeness of placenta; retained tissue can cause hemorrhage.
- Stage 4: monitor fundus, lochia, vitals, bladder, perineum, pain and anesthesia recovery; support bonding, skin-to-skin and feeding.
- Use respectful, trauma-informed care; explain procedures, obtain consent, protect privacy and include chosen support persons.

EMTALA

Emergency Medical Treatment and Labor Act: Medicare-participating hospitals must provide an appropriate medical screening exam for patients seeking emergency care, including patients with contractions, and stabilizing treatment within capability when an emergency condition exists. Transfers must meet applicable requirements. Do not delay screening or stabilization for insurance or payment questions.

5. GENERAL LABOR AND BIRTH COMPLICATIONS

- Risk factors include previous preterm birth, multiple gestation, hypertension/preeclampsia, diabetes, infection, abnormal placenta/presentation, obesity, uterine surgery, fluid abnormalities, substance use, anemia, and barriers to prenatal care.
- Nursing role: early recognition, maternal/fetal assessment, corrective interventions, rapid provider communication, emergency preparation, accurate documentation and family support.

6. PRETERM LABOR AND BIRTH

Definition and classification

What it is: Labor with contractions and cervical change before 37 weeks; preterm birth occurs before 37 weeks. Extremely preterm <28 weeks; very preterm 28 to <32; moderate/late 32 to <37.

Causes / risk factors: Previous preterm birth, short cervix, multiple gestation, genital/urinary infection, PPROM, overdistention, bleeding, smoking/substance use, uterine/cervical anomalies; often no clear cause.

Assessment and management: Recognize regular contractions, menstrual cramps, backache, pelvic pressure, discharge change, spotting or fluid leakage. Assess gestational age, maternal vitals, FHR/contractions, membranes, infection/bleeding, cervical assessment and ultrasound as ordered.

Medication	Use and nursing cautions
Betamethasone / dexamethasone	Antenatal steroids improve fetal lung maturity; commonly given at 24–33+6 weeks if delivery likely within 7 days, with selected use outside this range. Monitor maternal glucose.
Nifedipine	Tocolysis to delay birth briefly; monitor hypotension and symptoms.
Indomethacin	Selected early-gestation tocolysis; fetal ductal and renal risks with prolonged/later use.
Magnesium sulfate	Fetal neuroprotection when early delivery anticipated, commonly <32 weeks. Monitor reflexes, respirations, urine output. Antidote: calcium gluconate.
Penicillin G	Intrapartum GBS prophylaxis when indicated; verify allergies.

- Tocolytics typically aim to delay delivery around 48 hours for steroids/transfer, not indefinitely.
- Avoid tocolysis when infection, significant abruption, severe maternal/fetal compromise, or other indication for delivery exists.
- Magnesium toxicity: absent DTRs, respiratory depression, oliguria, weakness or altered consciousness → stop infusion, assess airway, summon help, prepare calcium gluconate.

7. PROM AND PPROM

Definition and causes

What it is: PROM = membrane rupture before labor; term PROM at ≥37 weeks. PPROM = preterm prelabor rupture before 37 weeks.

Causes / risk factors: Infection, prior PPROM, short cervix, twins, polyhydramnios, smoking, bleeding, membrane weakening.

Assessment and management: Sudden gush or persistent watery leakage; assess FHR immediately, time of rupture, fluid color/amount/odor, maternal temperature, infection signs, cord prolapse; avoid unnecessary digital exams.

Fluid	Meaning
Clear/pale straw	Typical fluid appearance.

Fluid	Meaning
Green/brown	Meconium-stained fluid.
Foul odor	Possible infection.
Bloody	Evaluate for placental complication or bleeding.

- Term PROM: generally plan delivery rather than prolonged expectant management; induction often used; GBS antibiotics when indicated.
- PPROM <34 weeks: if stable and no contraindications, expectant hospital management, fetal surveillance, infection monitoring, antenatal steroids, latency antibiotics, magnesium neuroprotection at appropriate gestation; deliver for infection, abruption, fetal compromise or other indication.
- PPROM 34–36+6 weeks: individualized delivery versus expectant care.
- Sudden FHR deceleration after rupture → evaluate for cord prolapse.

8. CHORIOAMNIONITIS / INTRA-AMNIOTIC INFECTION

Definition and causes

What it is: Infection/inflammation involving amniotic fluid, fetal membranes, placenta or intrauterine environment.

Causes / risk factors: Usually ascending genital-tract organisms; prolonged ROM/labor, multiple vaginal exams after ROM, internal monitoring, some genital infections.

Assessment and management: Maternal fever, tachycardia, fetal tachycardia, uterine tenderness, purulent/foul fluid, sometimes leukocytosis. Suspect with temp ≥39°C or 38–38.9°C plus another clinical risk factor, using current diagnostic criteria. Notify provider; IV broad-spectrum antibiotics (often ampicillin + gentamicin), monitor temperature/FHR, fluids/antipyretics as indicated, plan delivery by obstetric indications, notify neonatal team.

EXAM PRIORITY: Chorioamnionitis alone does not automatically require cesarean birth. Do not suppress labor when infection makes delivery necessary.

9. POSTTERM PREGNANCY, LABOR, AND BIRTH

Definition and causes

What it is: Late term 41+0 to 41+6 weeks; postterm ≥42 weeks.

Causes / risk factors: Often unknown; prior postterm pregnancy, first pregnancy, obesity, inaccurate dates, genetic/hormonal influences.

Assessment and management: Risks: oligohydramnios, cord compression, meconium/aspiration, macrosomia, shoulder dystocia, trauma, placental insufficiency, stillbirth. Postmaturity newborn may have peeling skin, long nails, reduced fat, wasted appearance, meconium stain. Confirm dates, fetal surveillance and fluid assessment, movement monitoring, discuss induction; generally recommend delivery by 42 weeks, may offer at 41 weeks.

10. DYSFUNCTIONAL LABOR (DYSTOCIA)

Definition and causes

What it is: Slow or arrested labor progress due to inadequate contractions, malposition, fetal-pelvic mismatch or other factors.

Causes / risk factors: Ineffective contractions, fetal malposition, maternal exhaustion, disproportion, overdistention. Hypotonic contractions are weak/infrequent; hypertonic or uncoordinated activity can be painful without effective progress.

Assessment and management: Assess contractions, FHR, dilation/station, fetal position, bladder and hydration. Encourage positioning, voiding, rest; consider ordered amniotomy/oxytocin if appropriate; monitor tachysystole; prepare operative delivery if needed.

- Protraction: slower-than-expected progress. Active-phase arrest criteria generally require ≥6 cm and ruptured membranes, no change after 4 hours adequate contractions OR 6 hours inadequate contractions with oxytocin, if safe to continue.
- Arrest of descent: second-stage descent stops; consider pushing time, parity, epidural, fetal position and clinical condition.
- Do not reflexively increase oxytocin without evaluating fetal tolerance and contraction adequacy.

11. OBESITY COMPLICATIONS

- Increased risks: gestational diabetes, hypertensive disorders, VTE, sleep apnea, macrosomia, dystocia, cesarean, anesthesia problems, infection and hemorrhage.
- Nursing care: correct-size BP cuff, safe positioning/handling, airway/anesthesia assessment, glucose/BP monitoring, mobility and VTE prevention, postoperative wound and bleeding checks, non-stigmatizing communication.

12. OBSTETRIC PROCEDURES

External cephalic version (ECV)

- Manual abdominal maneuver by clinician to turn breech/transverse fetus cephalic, usually offered near 37 weeks; may reduce cesarean for breech.
- Before: ultrasound confirmation, consent, FHR, emergency cesarean capability, assess Rh; terbutaline may be prescribed. During/after: FHR, contractions, bleeding, membranes, maternal discomfort. Give RhIG to eligible unsensitized Rh-negative patients as indicated.
- Risks: transient FHR changes, abruption, ROM, fetomaternal hemorrhage, rare emergency cesarean. Contraindications include placenta previa or other reasons vaginal birth is unsafe.

Induction vs. augmentation

Induction	Augmentation
Artificially initiates labor before spontaneous onset. Indications: postterm, PROM, hypertension or other maternal/fetal reasons.	Strengthens labor that already began spontaneously but is progressing inadequately.
Methods: cervical ripening, oxytocin, amniotomy.	Methods: oxytocin and/or amniotomy.
Avoid when vaginal birth contraindicated (e.g., placenta/vasa previa, persistent transverse lie, certain uterine scars, active genital HSV lesions/prodrome).	Assess fetal position, contraction pattern, progress and fetal tolerance before increasing stimulation.

Bishop score and cervical ripening

- Bishop score: dilation, effacement, consistency, position and station; score ≥8 commonly favorable (cutoffs vary).
- Dinoprostone (PGE2) and misoprostol (PGE1 analogue): ripen cervix and may cause contractions; monitor FHR/tachysystole.
- Foley or double-balloon catheter: mechanical ripening; monitor bleeding, discomfort and membrane status.
- Avoid misoprostol for term induction in patients with prior cesarean or major uterine surgery because of rupture risk; mechanical approaches may be considered.

Oxytocin (Pitocin) — HIGH ALERT

- What: synthetic uterotonic stimulates contractions. Why: induction, augmentation, prevention/treatment of postpartum uterine atony.
- Before: confirm indication, order, contraindications, baseline FHR and contraction pattern, IV access; use infusion pump/protocol.
- During: monitor FHR, frequency/duration/resting tone, BP/pulse, fluid balance; titrate and document per protocol.
- Risks: tachysystole, fetal hypoxemia/abnormal FHR, uterine rupture (especially scarred uterus), water intoxication/hyponatremia with prolonged high doses/excess fluids, hypotension.
- If tachysystole or fetal intolerance: stop/decrease oxytocin, lateral reposition, assess FHR/vitals, notify provider, give indicated fluids or prescribed terbutaline; do not increase dose.

Amniotomy

Artificial rupture of membranes with sterile instrument. Assess FHR immediately before and after, because of cord prolapse/compression risk; document fluid color, odor and amount.

Operative vaginal birth: forceps vs. vacuum

- Indications: prolonged second stage, nonreassuring fetal status when vaginal birth feasible, maternal exhaustion or medical need to limit pushing.
- Prerequisites: complete dilation, ruptured membranes, engaged head, known fetal position, appropriate clinical assessment, skilled clinician.
- Forceps: assist head delivery/selected rotation; risks maternal perineal/cervical trauma, fetal facial/nerve injury, rare skull injury.

- Vacuum: scalp suction cup; risks scalp injury, cephalohematoma, subgaleal hemorrhage, jaundice. Generally avoid vacuum before 34 weeks.
- After vacuum: assess increasing scalp swelling, pallor, tachycardia and shock; subgaleal hemorrhage can be life-threatening.

Cesarean delivery

- Surgical delivery through abdomen/uterus. Indications include previa, abnormal lie, persistent fetal compromise, labor arrest, rupture, cord prolapse when immediate vaginal birth not feasible, some uterine scars.
- Pre-op: consent, FHR/vitals, IV/labs, prophylactic antibiotics, anesthesia preparation, urinary catheter when indicated, surgical checks.
- Post-op: vitals, bleeding/fundus, incision, pain, urine output, mobility/VTE prevention, infection surveillance, bonding/feeding.
- Risks: hemorrhage, infection, VTE, organ injury, anesthesia and wound complications.

TOLAC and VBAC

- TOLAC = trial of labor after cesarean; VBAC = successful vaginal birth after cesarean. Many patients with one prior low-transverse uterine incision may qualify depending on history/resources.
- Contraindications often include prior classical uterine incision, prior uterine rupture, certain extensive uterine surgeries, or current contraindication to vaginal birth.
- Nursing: continuous FHR monitoring, labor progress and rupture surveillance, immediate emergency capability.
- Uterine rupture warning: sudden abnormal FHR/bradycardia (often earliest), pain, bleeding, loss of fetal station, maternal hypotension or altered contractions.

13. OBSTETRIC EMERGENCIES

Umbilical cord prolapse

What it is: Cord falls below/alongside presenting part and becomes compressed.

Causes / risk factors: Unengaged presenting part with ROM, malpresentation, prematurity, polyhydramnios, multiples.

Assessment and management: Sudden bradycardia/severe variables; cord may be palpable/visible. CALL FOR HELP; sterile-gloved hand elevates presenting part OFF cord and stays there; knee-chest or steep Trendelenburg as appropriate; stop oxytocin; emergency delivery, usually cesarean. Never push cord back or release elevation prematurely.

Shoulder dystocia

What it is: Shoulders become impacted after head delivers; anterior shoulder often behind pubic symphysis.

Causes / risk factors: Macrosomia, diabetes, obesity, prior dystocia, operative vaginal birth; can occur without risks.

Assessment and management: Turtle sign (head retracts). Call for help, time event, assist McRoberts (knees toward abdomen) and suprapubic pressure, prepare other maneuvers/neonatal resuscitation, assess maternal trauma/bleeding. NEVER fundal pressure. Risks brachial plexus injury, fracture, hypoxia, severe lacerations, hemorrhage.

Placental abruption

What it is: Placenta separates prematurely from uterine wall.

Causes / risk factors: Hypertension/preeclampsia, trauma, cocaine, smoking, previous abruption.

Assessment and management: Painful bleeding (may be concealed), rigid tender uterus, frequent contractions, fetal distress, shock. Activate response, maternal stabilization, continuous FHR, large-bore IV, CBC/coags/fibrinogen/type-cross, blood products and urgent delivery as indicated; monitor DIC.

Placenta previa

What it is: Placenta covers internal cervical opening.

Causes / risk factors: Placental implantation abnormality; history of uterine surgery may increase risk.

Assessment and management: Classically painless bright-red bleeding, soft nontender uterus. Assess mother/fetus, IV, ultrasound, monitor bleeding, prepare cesarean as indicated. NO digital vaginal exam until location known.

Uterine rupture

What it is: Full-thickness disruption of uterine wall.

Causes / risk factors: Prior uterine scar, TOLAC, previous rupture, excessive stimulation, certain uterine operations.

Assessment and management: Sudden fetal bradycardia, possible pain/bleeding, loss of station, hypotension, changed contractions. Activate emergency response, stop oxytocin, support ABCs, large-bore IV/blood, immediate surgery and neonatal resuscitation.

Amniotic fluid embolism

What it is: Rare catastrophic maternal inflammatory/cardiopulmonary reaction associated with amniotic fluid components.

Causes / risk factors: Often unpredictable; can occur during labor or soon after birth.

Assessment and management: Sudden hypoxia, respiratory distress, hypotension/collapse, seizures, DIC. Activate resuscitation, airway/oxygenation, CPR if arrest, blood products/coagulopathy treatment, critical care and delivery when indicated.

Postpartum hemorrhage (PPH)

What it is: ≥1000 mL cumulative blood loss or bleeding with hypovolemia signs within 24 hours, any delivery route; intervene sooner for instability.

Causes / risk factors: Four Ts: Tone (atony, most common), Trauma (lacerations/hematoma/rupture), Tissue (retained placenta), Thrombin (coagulopathy).

Assessment and management: Boggy uterus + heavy bleeding: call for help, fundal massage, quantify loss, vitals, uterotonics, large-bore IV, fluids/blood, empty bladder, additional procedures if persistent.

PPH medication	Key precaution
Oxytocin	First-line uterotonic; monitor hemodynamics/fluid balance.
Methylergonovine (Methergine)	Avoid with hypertension/preeclampsia.

PPH medication	Key precaution
Carboprost (Hemabate)	Avoid with asthma (bronchospasm).
Misoprostol	Uterotonic; fever, chills and diarrhea possible.
Tranexamic acid (TXA)	Antifibrinolytic, NOT uterotonic; prompt treatment, generally within 3 hours of birth.

Eclampsia

What it is: New seizures associated with hypertensive disorder of pregnancy when other cause not identified.

Causes / risk factors: Preeclampsia; warning signs severe headache, visual changes, RUQ/epigastric pain, severe BP, clonus, altered mental status.

Assessment and management: Call help, protect from injury, lateral position, airway/suction/O2 as indicated, DO NOT restrain or place objects in mouth, magnesium sulfate for seizures, monitor RR/DTR/urine/FHR, treat severe BP with prescribed IV labetalol/hydralazine or oral immediate-release nifedipine; delivery after stabilization as indicated.

EXAM PRIORITY: Magnesium sulfate treats/prevents eclamptic seizures but is NOT the primary antihypertensive.

14. RAPID EMERGENCY RECOGNITION

Clue	Likely emergency	Priority response
Bradycardia after ROM + cord palpable	Cord prolapse	Lift presenting part off cord, call emergency team.
Turtle sign	Shoulder dystocia	Call help; McRoberts and suprapubic pressure.
Painful bleeding + rigid uterus	Abruption	Stabilize, continuous FHR, prepare blood/delivery.
Painless bright-red bleeding	Previa	NO digital exam; ultrasound and assess.
TOLAC + bradycardia/loss of station	Uterine rupture	Emergency surgery response.
Sudden hypoxia + hypotension + DIC	Amniotic fluid embolism	Cardiopulmonary resuscitation support.
Boggy uterus + heavy bleeding	Uterine atony	Fundal massage, hemorrhage protocol.
Seizure with preeclampsia	Eclampsia	Protect airway, magnesium.
Oxytocin + tachysystole + late decels	Excess uterine stimulation	Stop oxytocin, reposition, correct causes.
Fever + fetal tachycardia after prolonged ROM	Intra-amniotic infection	Notify, IV antibiotics, monitor.

15. NCLEX-STYLE PRACTICE QUESTIONS + RATIONALES

Question 1

A patient is 7 cm/90%/−1. What does −1 mean?

Answer: The presenting part is above the ischial spines. Rationale: Negative station means above the ischial spines.

Question 2

Oxytocin is infusing; recurrent late decelerations and six contractions/10 minutes occur. Priority?

Answer: Stop oxytocin and reposition. Rationale: Tachysystole with fetal intolerance requires stopping stimulation and corrective measures.

Question 3

Which FHR finding is most reassuring?

Answer: Moderate variability. Rationale: 6–25 bpm variability indicates reassuring autonomic response.

Question 4

At 30 weeks, which medication may provide fetal neuroprotection?

Answer: Magnesium sulfate. Rationale: Used when early preterm delivery is anticipated.

Question 5

Most important immediately after amniotomy?

Answer: Assess fetal heart rate. Rationale: Detect cord prolapse or compression.

Question 6

Boggy uterus with heavy postpartum bleeding: first action?

Answer: Fundal massage and call for help. Rationale: Uterine atony is a hemorrhage emergency.

Question 7

Which PPH medication should be questioned with asthma?

Answer: Carboprost. Rationale: Can trigger bronchospasm.

Question 8

TOLAC with sudden fetal bradycardia and loss of station: suspect?

Answer: Uterine rupture. Rationale: Classic warning signs.

Question 9

Head retracts against perineum after birth: anticipate?

Answer: McRoberts maneuver. Rationale: Turtle sign indicates shoulder dystocia; avoid fundal pressure.

Question 10

Suspected placenta previa: which order should be questioned?

Answer: Digital cervical exam. Rationale: May precipitate severe bleeding.

Question 11

Which finding suggests magnesium toxicity?

Answer: Absent reflexes with respiratory depression. Rationale: Stop infusion, summon help, prepare calcium gluconate.

Question 12

Fever 39°C + fetal tachycardia + prolonged ROM: suspect?

Answer: Chorioamnionitis. Rationale: Suggests intra-amniotic infection requiring antibiotics and monitoring.

16. LAST-MINUTE MEMORIZATION SHEET

Topic	Must know
FHR baseline	110–160 bpm
Moderate variability	6–25 bpm
Acceleration ≥32 weeks	15 bpm × 15 seconds
Acceleration <32 weeks	10 bpm × 10 seconds
Prolonged deceleration	2 to <10 minutes
Tachysystole	>5 contractions/10 min averaged over 30 min
Active labor / complete dilation	~6 cm onward / 10 cm
Preterm / postterm	<37 weeks / ≥42 weeks
Magnesium antidote	Calcium gluconate
Methergine / Hemabate	Avoid in hypertension / asthma
Cord prolapse / shoulder dystocia	Elevate presenting part / McRoberts + suprapubic pressure
Uterine atony	Fundal massage + uterotonics
Late / variable decelerations	Uteroplacental perfusion / cord compression
Epidural hypotension	Lateral position, support circulation, notify anesthesia
Uterine rupture	Sudden fetal bradycardia, especially during TOLAC
Chorioamnionitis	Fever + fetal tachycardia → IV antibiotics

Final nursing priorities

- Maternal airway, breathing and circulation support fetal oxygenation.
- Interpret FHR in context: baseline, variability, decelerations, contractions and maternal status.

- Oxytocin is high alert; stop/reduce when tachysystole or fetal intolerance occurs.
- Avoid unsafe vaginal examinations with suspected previa or PPROM.
- Induction starts labor; augmentation strengthens labor already underway.
- Recognize and escalate cord prolapse, shoulder dystocia, rupture, abruption and collapse immediately.
- Know drug precautions: Methergine/hypertension; Hemabate/asthma; misoprostol induction/prior cesarean.

Educational study material. Clinical decisions must follow current institutional protocols, provider orders, and applicable obstetric guidelines. The practice questions reproduce the concepts and rationales from the study guide in print-friendly form.